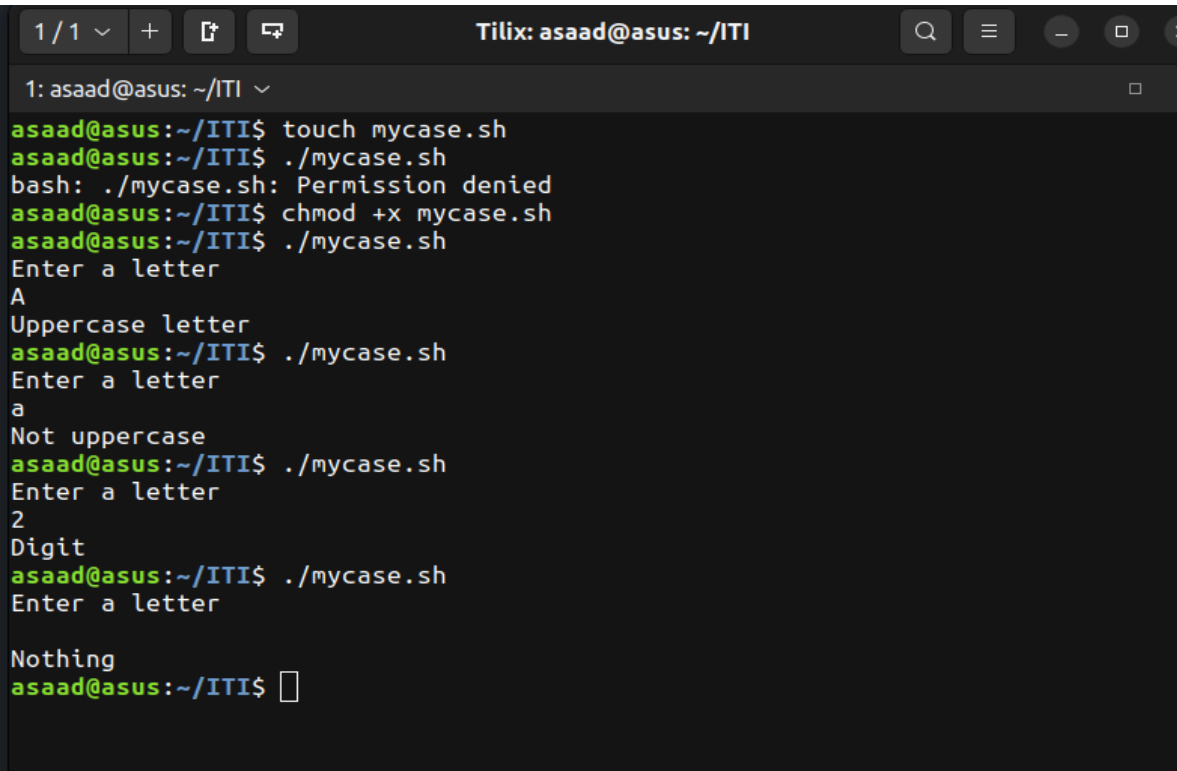


1. Write a script called mycase, using the case utility to checks the type of character entered by a user:
 - a. Upper Case.
 - b. Lower Case.
 - c. Number.
 - d. Nothing.

```
#!/usr/bin/bash

echo "Enter a letter"
read REPLY

if [[ $REPLY =~ [A-Z] ]]; then
    echo "Uppercase letter"
elif [[ $REPLY =~ [a-z] ]]; then
    echo "Not uppercase"
elif [[ $REPLY =~ [0-9] ]]; then
    echo "Digit"
else
    echo "Nothing"
fi
```



The terminal window shows the following sequence of commands and outputs:

```
Tilix: asaad@asus: ~/ITI
1: asaad@asus: ~/ITI
asaad@asus:~/ITI$ touch mycase.sh
asaad@asus:~/ITI$ ./mycase.sh
bash: ./mycase.sh: Permission denied
asaad@asus:~/ITI$ chmod +x mycase.sh
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
A
Uppercase letter
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
a
Not uppercase
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
2
Digit
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
Nothing
asaad@asus:~/ITI$
```

2. Enhanced the previous script, by checking the type of string entered by a user:

- a. Upper Cases.
- b. Lower Cases.
- c. Numbers.
- d. Mix.
- e. Nothing.

```
#!/usr/bin/bash

echo "Enter a letter"
read REPLY

if [[ $REPLY =~ ^[A-Z]+$ ]]; then
    echo "Uppercase"
elif [[ $REPLY =~ ^[a-z]+$ ]]; then
    echo "Lowercase"
elif [[ $REPLY =~ ^[0-9]+$ ]]; then
    echo "Numbers"
elif [[ $REPLY =~ ^[A-Za-z0-9]+$ ]]; then
    echo "Mix"
else
    echo "Nothing"
fi
```

```
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
aA
Mix
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
as12
Mix
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
a
Lowercase
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
A
Uppercase
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter

Nothing
asaad@asus:~/ITI$ ./mycase.sh
Enter a letter
12
Numbers
```

3. Write a script called mychmod using for utility to give execute permission to all files and directories in your home directory.

```
#!/usr/bin/bash
```

```
for file in ~/*; do
    chmod +x "$file"
done
```

```
asaad@asus:~/ITI$ touch mychmod.sh
asaad@asus:~/ITI$ chmod +x mychmod.sh
asaad@asus:~/ITI$ ./mychmod.sh
```

```
asaad@asus:~/ITI$ ls -l ~
```

```
total 32732
drwxrwxr-x 9 asaad asaad 4096 Jan 31 15:23 AngularPortfolio
d--x--x--x 2 asaad asaad 4096 Oct 27 15:43 asa
drwxr-xr-x 3 asaad asaad 4096 Jul 9 2020 aws
-rwxrwxr-x 1 asaad asaad 33095846 Jan 14 00:30 awscliv2.zip
drwxrwxr-x 3 asaad asaad 4096 Feb 18 17:46 book-store-frontend
drwxr-xr-x 4 asaad asaad 4096 Aug 26 2025 bootstrap-5.3.8-dist
drwxr-xr-x 2 asaad asaad 4096 Feb 14 23:06 Desktop
drwxr-xr-x 12 asaad asaad 4096 Feb 20 11:25 Documents
drwxr-xr-x 2 asaad asaad 4096 Feb 28 14:39 Downloads
drwxrwxr-x 4 asaad asaad 4096 Feb 22 13:58 ECommerceAngularNode
drwxr-xr-x 6 asaad asaad 4096 Dec 13 22:45 'E COMMERCE PROJECT'
```

4. Write a script called mybackup using for utility to create a backup of only files in your home directory.

```
#!/usr/bin/bash
```

```
mkdir -p ~/backup
for file in ~/*; do
    if [ -f "$file" ]; then
        cp "$file" ~/backup/
    fi
done
echo "Backup completed"
```

```
asaad@asus:~/ITI$ touch backup.sh
asaad@asus:~/ITI$ chmod +x backup.sh
asaad@asus:~/ITI$ ./backup.sh
```

```
Backup completed
```

```
asaad@asus:~/ITI$ ls
```

```
'AAConvia111111 (2).pdf'  fitnessApp  Networks
backup.sh               forkify     randomQuoteGenerator
Basket                  github-recovery-codes.txt  RH
'c-c++ projects'        java        weatherApp
chickenInvadersJs      mycase.sh
cpp2.md                 mychmod.sh
```

```
asaad@asus:~/ITI$ cd ..
```

```
asaad@asus:~$ cd backup/
```

```
asaad@asus:~/backup$ ls
```

```
awscliv2.zip      mycp2.sh  parameters.java  test.sh
govtech_orchestrator.pdf  mycp3.sh  root.java        'Untitled Document 1'
hellouser.sh      mycp.sh   s1.sh            'Untitled Document 155'
hi1.sh            myls.sh   s2.sh
```

```
asaad@asus:~/backup$
```

5. Write a script called mymail using for utility to send a mail to all users in the system. Note: write the mail body in a file called mtemplate.

```
#!/usr/bin/bash
for user in $(cut -d: -f1 /etc/passwd); do
    mail -s "System Message" "$user" < mtemplate
done
echo "Mail sent to all users"
```

6. Write a script called chkmail to check for new mails every 10 seconds. Note: mails are saved in /var/mail/username.

```
#!/usr/bin/bash
mail="/var/mail/$USER"
while true; do
    if [ -s "$mail" ]; then
        echo "new mail"
    else
        echo "no new mail"
    fi
    sleep 10
done
```